MSC 201 **98421**

Unit 1 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A database codes printer brands as 1, 2, and 3. Why would averaging these codes be misleading?	A. The numbers are labels; recoding them could change the mean. B. Every numerical entry measures an amount. C. Brand codes cannot be counted. D. The largest code is always the most common brand. 10 pts
2. Digital purchases rise from 40 of 100 to 60 of 200. What happened?	A. The count rose, but the share fell. B. The count fell, but the share rose. C. Both count and share rose. D. Both count and share fell. 10 pts
3. Four jobs have a mean of 5 pages. What is the total number of pages?	A. 20. B. 1.25. C. 9. D. 5. 10 pts
4. Remove 100 from 1, 2, 3, 4, 100. What happens to the mean and median?	A. The median increases and mean stays unchanged. B. The mean increases and median decreases. C. Both decrease, with a much larger decrease in the mean. D. Both stay unchanged. 10 pts
5. A systematic sample starts at ID 3 and takes every tenth ID. Which IDs appear first?	A. 10, 20, 30. B. 3, 13, 23. C. 3, 6, 9. D. 3, 10, 20. 10 pts
6. A score is reported at approximately the 90th percentile. What kind of information does that give?	A. A score of exactly 90 points. B. Its position relative to other scores. C. The class mean being 90. D. That 90% of exam answers were correct. 10 pts

Test continues on the other side of the sheet.

MSC 201 **98421**

Unit 1 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. One teacher uses format A and another uses format B. Why is a score difference hard to attribute?	A. There are too many treatment groups to compare. B. Teacher and format effects are mixed together. C. Different teachers guarantee format A is better. D. Scores cannot be numerical. 10 pts
8. A sample has 8 observations. Its sum of squared deviations is already supplied. Which divisor gives sample variance?	A. 8. B. The sample mean. C. 64. D. 7. 10 pts
9. Categories in a bar graph are reordered alphabetically. What changes?	A. Every relative frequency. B. The number of observations in each category. C. The most frequent category. D. The display order, but not category frequencies. 10 pts
10. A table has positive frequency at exactly 2 jobs. How do counts for fewer than 2 and at most 2 compare?	A. At most 2 has the larger count. B. Neither can be found from exact frequencies. C. They must be equal. D. Fewer than 2 has the larger count. 10 pts

98421

TEST PAGE 1

1. A.

10 pts

2. A.

10 pts

3. A.

10 pts

4. C.

10 pts

5. B.

10 pts

6. B.

10 pts

TEST PAGE 2

7. B.

10 pts

8. D.

10 pts

9. D.

10 pts

10. A.

10 pts

MSC 201 **59420**

Unit 1 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. For 2, 3, 3, 4, 18, how does the mean compare with the median?	A. The mean is smaller. B. They are equal. C. No comparison is possible without a calculator. D. The mean is larger. 10 pts
2. A researcher deletes genuine results that disagree with the desired conclusion. What is the problem?	A. Any unusual observation is automatically an error. B. Deleting inconvenient values always improves accuracy. C. The report no longer fairly represents the collected evidence. D. A smaller dataset guarantees less bias. 10 pts
3. Two properly drawn random samples give support rates of 42% and 47%. Which explanation is reasonable?	A. Different rates prove one survey was fabricated. B. The larger percentage must be the true population rate. C. Chance sampling variation can produce different rates. D. Random samples always have equal percentages. 10 pts
4. A systematic sample starts at ID 3 and takes every tenth ID. Which IDs appear first?	A. 3, 10, 20. B. 3, 6, 9. C. 10, 20, 30. D. 3, 13, 23. 10 pts
5. A population contains 4 observations with a sum of squared deviations from its mean of 36. What is the population variance?	A. 36. B. 9. C. 3. D. 12. 10 pts
6. Using the reported quartiles, group A has Q1=3, Q3=9; group B has Q1=10, Q3=14. Which has greater IQR?	A. Group A. B. Group B. C. The larger median must determine the IQR. D. They are equal. 10 pts

Test continues on the other side of the sheet.

MSC 201 **59420**

Unit 1 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. Service A has 10 complaints in 100 jobs; B has 20 in 400 jobs. Which has the lower complaint rate?	A. The service with fewer complaints must have the lower rate. B. Service A. C. They have equal rates. D. Service B. 10 pts
8. A histogram is rebuilt using wider bins. What is likely lost?	A. The numerical nature of the variable. B. The need for an axis scale. C. The total number of observations. D. Fine detail such as small gaps and local peaks. 10 pts
9. A bell-shaped distribution of scores has mean 50 and standard deviation 5. Approximately what percentage of scores lies between 45 and 55?	A. 95%. B. 68%. C. 99.7%. D. 50%. 10 pts
10. A recorded file size is far above the rest. What should happen before deleting it?	A. Delete it because all outliers are mistakes. B. Check whether it is an error or a genuine unusual file. C. Replace it with zero automatically. D. Keep it only if the mean equals the median. 10 pts

59420

TEST PAGE 1

1.

D.

10 pts

2.

C.

10 pts

3.

C.

10 pts

4.

D.

10 pts

5.

B.

10 pts

6. A.

10 pts

TEST PAGE 2

7.

D.

10 pts

8.

D.

10 pts

9.

B.

10 pts

10.

B.

10 pts

MSC 201 **14012**

Unit 1 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Using the reported quartiles, group A has Q1=3, Q3=9; group B has Q1=10, Q3=14. Which has greater IQR?	A. Group A. B. They are equal. C. Group B. D. The larger median must determine the IQR. 10 pts
2. Someone claims mean > median is a universal proof that any dataset is right-skewed. What is the best response?	A. It proves every observation exceeds the median. B. It is a useful common pattern, not a universal characterization. C. It proves the dataset is normal. D. It guarantees exactly one mode. 10 pts
3. A dataset's standard deviation is zero. What must be true?	A. There is one high outlier. B. The mean is zero. C. All observations are equal. D. All observations are zero. 10 pts
4. Students are randomly invited, but only 15% respond. Which concern remains?	A. The invitations were necessarily nonrandom. B. Random invitation guarantees representative responses. C. Respondents may differ from nonrespondents. D. Low response proves every answer false. 10 pts
5. A bar graph of counts 98 and 100 starts its vertical axis at 97. What impression might it exaggerate?	A. The total number of observations. B. The number of categories. C. The size of the difference between the counts. D. The fact that 98 is below 100. 10 pts
6. Exact wait times are replaced by short, medium, and long categories. What information is lost?	A. The number of observations. B. Whether any observations exist. C. The ordering of the three categories. D. The exact differences between wait times. 10 pts

Test continues on the other side of the sheet.

MSC 201 **14012**

Unit 1 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. Randomly assigned groups have slightly different initial mean scores. What follows?	A. Random assignment requires exact matching of every score. B. The assignment was necessarily dishonest. C. The study must have used a census. D. Chance imbalance can occur despite random assignment. 10 pts
8. Most files are small, with a few extremely large files. Which better describes a typical file's size?	A. The median. B. The maximum. C. The total size alone. D. The mean student ID. 10 pts
9. A room at 20 degrees Celsius is compared with one at 10 degrees Celsius. Which statement is justified?	A. The first is twice as comfortable for every person. B. The first is 10 degrees warmer on that scale. C. Celsius readings cannot be subtracted. D. The first has twice the absolute temperature. 10 pts
10. A box plot flags an unusually large print job. Which conclusion is warranted?	A. The job must be fraudulent. B. The rest of the data must be normal. C. The job deserves investigation; the plot alone does not prove error. D. The value must be deleted before any summary. 10 pts

14012

TEST PAGE 1

1. A.

10 pts

2. B.

10 pts

3. C.

10 pts

4. C.

10 pts

5. C.

10 pts

6. D.

10 pts

TEST PAGE 2

7. D.

10 pts

8. A.

10 pts

9. B.

10 pts

10. C.

10 pts

MSC 201 **68191**

Unit 1 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A population contains 4 observations with a sum of squared deviations from its mean of 36. What is the population variance?	A. 36. B. 12. C. 9. D. 3. 10 pts
2. Students rate a service while its manager watches. Compared with anonymous responses, what might happen?	A. Every rating must become negative. B. Students may avoid negative ratings. C. The manager's presence guarantees accuracy. D. Sample size automatically doubles. 10 pts
3. Categories in a bar graph are reordered alphabetically. What changes?	A. Every relative frequency. B. The most frequent category. C. The number of observations in each category. D. The display order, but not category frequencies. 10 pts
4. Students choosing tutoring score higher. Why is a causal claim uncertain?	A. Any score difference proves a tutoring effect. B. Motivation may affect both tutoring attendance and scores. C. A larger score difference removes all confounding. D. Tutoring can never help students. 10 pts
5. Two datasets have identical five-number summaries. What follows?	A. They must contain identical observations. B. Their means must be identical. C. Their sample sizes must be identical. D. They can still have different internal distributions. 10 pts
6. A sample's mean file size is 18 KB. The study aims to estimate the mean across all files on a server. What does 18 KB represent?	A. The size of every file on the server. B. An estimate based on the sampled files. C. The exact population mean regardless of selection. D. A guaranteed maximum file size. 10 pts
7. A grouped table reports 12 durations from 5 to less than 10 seconds. Can it tell exactly how many equal 7 seconds?	A. Yes; none may equal 7. B. Yes; exactly half must equal 7. C. No; exact positions within the interval are hidden. D. Yes; all 12 must equal 7. 10 pts

Test continues on the other side of the sheet.

MSC 201 **68191**

Unit 1 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. Most files are small, with a few extremely large files. Which better describes a typical file's size?	A. The mean student ID. B. The median. C. The maximum. D. The total size alone. 10 pts
9. Data are 2, 4, 6, 8, 10. How do mean and median compare?	A. The mean exceeds the median. B. The median is the maximum. C. They are equal. D. The mean is below the median. 10 pts
10. A sample variance is 9 minutes squared. What is its standard deviation?	A. 81 minutes. B. 3 minutes squared. C. 9 minutes. D. 3 minutes. 10 pts

68191

TEST PAGE 1

1.

C.

10 pts

2.

B.

10 pts

3.

D.

10 pts

4.

B.

10 pts

5.

D.

10 pts

6.

B.

10 pts

7.

C.

10 pts

TEST PAGE 2

8.

B.

10 pts

9.

C.

10 pts

10.

D.

10 pts

MSC 201 **55371**

Unit 1 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. The ordered data are 2, 4, 4, 7, 20. What is the median?	A. 4. B. 6. C. 20. D. 7. 10 pts
2. Campus A has 30 users among 60 students; campus B has 40 among 100. Which has the higher use rate?	A. Campus A. B. Campus B. C. They have equal rates. D. Counts alone prove campus B has the higher rate. 10 pts
3. Frequencies for 0, 1, 2, and 3 jobs are 2, 3, 4, and 1. How many sessions had at most 1 job?	A. 5. B. 3. C. 9. D. 2. 10 pts
4. Two services have mean wait 10 minutes, but SDs 1 and 6 minutes. What does the larger SD suggest?	A. The mean must actually be 16 minutes. B. Waits vary more around the same mean. C. The median must be 6 minutes. D. Every wait is six minutes longer. 10 pts
5. A dataset has mean 10. One new observation equal to 10 is added. What happens to the mean?	A. It stays 10. B. It falls. C. It becomes zero. D. It rises. 10 pts
6. Plan A samples students from every building. Plan B randomly chooses two buildings and surveys everyone there. What distinguishes Plan B?	A. It selects some students from every group. B. It guarantees all possible student samples are equally likely. C. It necessarily includes every building. D. It studies all students in selected groups. 10 pts
7. Volunteers are randomly assigned to two practice formats. What does random assignment primarily help with?	A. Guaranteeing identical scores in both groups. B. Eliminating the need to measure outcomes. C. Ensuring volunteers represent all students. D. Balancing other influences between treatment groups. 10 pts

Test continues on the other side of the sheet.

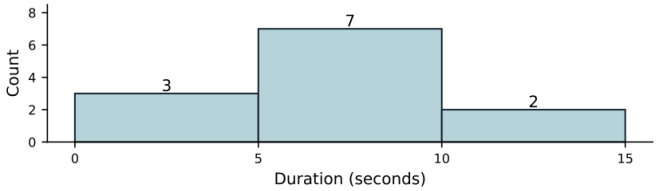
MSC 201 **55371**

Unit 1 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. Histogram counts are 3 in [0,5), 7 in [5,10), and 2 in [10,15). How many values are below 10? 	A. 10. B. 7. C. 12. D. 2. 10 pts
9. A dataset has mean 8. One observation below 8 is removed. What happens to the mean of the remaining values?	A. It increases. B. It becomes the removed value. C. It always stays 8. D. It decreases. 10 pts
10. Data are 0, 8, 9, 9, 10. Which center is lower?	A. The median. B. The maximum. C. The mean. D. They are equal. 10 pts

55371

TEST PAGE 1

1. A.

10 pts

2. A.

10 pts

3. A.

10 pts

4. B.

10 pts

5. A.

10 pts

6. D.

10 pts

7. D.

10 pts

TEST PAGE 2

8. A.

10 pts

9. A.

10 pts

10. C.

10 pts

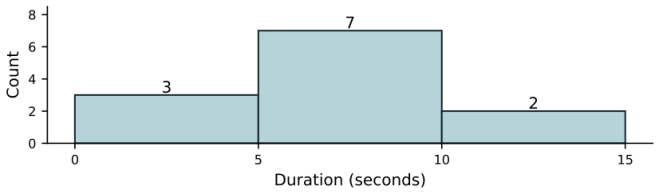
MSC 201 **50742**

Unit 1 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A room at 20 degrees Celsius is compared with one at 10 degrees Celsius. Which statement is justified?	A. The first is twice as comfortable for every person. B. Celsius readings cannot be subtracted. C. The first is 10 degrees warmer on that scale. D. The first has twice the absolute temperature. 10 pts
2. A college wants each academic program represented in a survey. Which plan best achieves that directly?	A. Randomly sample students within every program. B. Survey only the largest program. C. Select the first students entering one room. D. Invite only students who previously complained. 10 pts
3. Histogram counts are 3 in [0,5), 7 in [5,10), and 2 in [10,15). How many values are below 10? 	A. 10. B. 12. C. 7. D. 2. 10 pts
4. A file is 60 KB; its group has mean 50 KB and SD 5 KB. How far above the mean is it?	A. 2 standard deviations. B. 0.5 standard deviations. C. 10 standard deviations. D. 12 standard deviations. 10 pts
5. A mean is estimated from grouped intervals using midpoints. Why is it generally approximate?	A. The sample size is necessarily unknown. B. Actual observations need not equal the interval midpoints. C. Frequencies cannot be used in means. D. Midpoints are never numerical values. 10 pts
6. A random sample of 80 enrolled students reports 30% support for later hours. Which conclusion is justified?	A. Exactly 30% of all students support later hours. B. Exactly 24 students in the entire college support later hours. C. The population percentage must exceed 30%. D. 30% describes the sample and estimates population support. 10 pts

Test continues on the other side of the sheet.

MSC 201 **50742**

Unit 1 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. Two datasets have identical five-number summaries. What follows?	A. They must contain identical observations. B. Their sample sizes must be identical. C. Their means must be identical. D. They can still have different internal distributions. 10 pts
8. A public dataset includes names beside individual survey answers. Which change best protects privacy while preserving group summaries?	A. Replace every answer with the overall mean. B. Publish only the names. C. Keep names and omit the study methods. D. Remove identifying details before sharing. 10 pts
9. Someone claims mean > median is a universal proof that any dataset is right-skewed. What is the best response?	A. It guarantees exactly one mode. B. It proves the dataset is normal. C. It proves every observation exceeds the median. D. It is a useful common pattern, not a universal characterization. 10 pts
10. A box plot flags an unusually large print job. Which conclusion is warranted?	A. The rest of the data must be normal. B. The value must be deleted before any summary. C. The job deserves investigation; the plot alone does not prove error. D. The job must be fraudulent. 10 pts

50742

TEST PAGE 1

1.

C.

10 pts

2. A.

10 pts

3. A.

10 pts

4. A.

10 pts

5.

B.

10 pts

6.

D.

10 pts

TEST PAGE 2

7.

D.

10 pts

8.

D.

10 pts

9.

D.

10 pts

10.

C.

10 pts

MSC 201 **90636**

Unit 1 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Students are randomly invited, but only 15% respond. Which concern remains?	A. The invitations were necessarily nonrandom. B. Random invitation guarantees representative responses. C. Low response proves every answer false. D. Respondents may differ from nonrespondents. 10 pts
2. A box plot flags an unusually large print job. Which conclusion is warranted?	A. The rest of the data must be normal. B. The job must be fraudulent. C. The job deserves investigation; the plot alone does not prove error. D. The value must be deleted before any summary. 10 pts
3. Frequencies for 0, 1, 2, and 3 jobs are 2, 3, 4, and 1. How many sessions had at most 1 job?	A. 9. B. 3. C. 5. D. 2. 10 pts
4. In 2, 3, 3, 4, 18, the 18 is replaced by 180. What happens?	A. The mean decreases. B. Both stay unchanged. C. The mean increases; the median stays 3. D. The median increases more than the mean. 10 pts
5. A sample's mean file size is 18 KB. The study aims to estimate the mean across all files on a server. What does 18 KB represent?	A. The exact population mean regardless of selection. B. The size of every file on the server. C. An estimate based on the sampled files. D. A guaranteed maximum file size. 10 pts
6. Students choosing tutoring score higher. Why is a causal claim uncertain?	A. Any score difference proves a tutoring effect. B. A larger score difference removes all confounding. C. Tutoring can never help students. D. Motivation may affect both tutoring attendance and scores. 10 pts

Test continues on the other side of the sheet.

MSC 201 **90636**

Unit 1 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. Data are 0, 8, 9, 9, 10. Which center is lower?	A. The mean. B. They are equal. C. The maximum. D. The median. 10 pts
8. A bar graph of counts 98 and 100 starts its vertical axis at 97. What impression might it exaggerate?	A. The size of the difference between the counts. B. The total number of observations. C. The number of categories. D. The fact that 98 is below 100. 10 pts
9. Why is adding signed deviations from the sample mean unsuitable as a spread measure?	A. The mean is always the largest value. B. Positive and negative deviations cancel. C. All deviations are positive. D. It necessarily counts each value twice. 10 pts
10. Using the reported quartiles, group A has Q1=3, Q3=9; group B has Q1=10, Q3=14. Which has greater IQR?	A. They are equal. B. Group B. C. The larger median must determine the IQR. D. Group A. 10 pts

90636

TEST PAGE 1

1.

D.

10 pts

2.

C.

10 pts

3.

C.

10 pts

4.

C.

10 pts

5.

C.

10 pts

6.

D.

10 pts

TEST PAGE 2

7. A.

10 pts

8. A.

10 pts

9.

B.

10 pts

10.

D.

10 pts

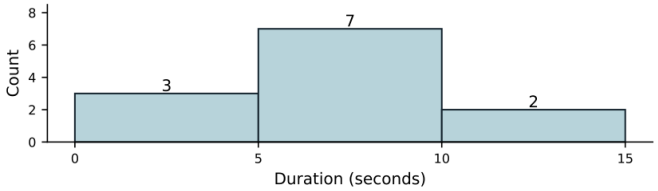
MSC 201 **92081**

Unit 1 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Histogram counts are 3 in [0,5), 7 in [5,10), and 2 in [10,15). How many values are below 10? 	A. 2. B. 7. C. 10. D. 12. 10 pts
2. A cumulative table gives 30% below 5 seconds and 75% below 10 seconds. What percent is from 5 to less than 10?	A. 30%. B. 105%. C. 75%. D. 45%. 10 pts
3. A printer study uses only one printer per paper type. Why is that weak evidence?	A. Two observations guarantee a representative result. B. Replication would eliminate all possible errors. C. Paper type cannot affect printer performance. D. Printer-specific differences may dominate the comparison. 10 pts
4. Two datasets have identical five-number summaries. What follows?	A. Their sample sizes must be identical. B. Their means must be identical. C. They must contain identical observations. D. They can still have different internal distributions. 10 pts
5. The numbers of pages in six jobs are 1, 2, 2, 4, 4, 5. Which statement about the mode is correct?	A. The mode is 3, halfway between 2 and 4. B. Only 4 is a mode. C. Only 2 is a mode. D. Both 2 and 4 are modes. 10 pts
6. A sample's mean file size is 18 KB. The study aims to estimate the mean across all files on a server. What does 18 KB represent?	A. A guaranteed maximum file size. B. The exact population mean regardless of selection. C. An estimate based on the sampled files. D. The size of every file on the server. 10 pts

Test continues on the other side of the sheet.

MSC 201 **92081**

Unit 1 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. Notebook colors are blue, blue, black, red. Which statement meaningfully summarizes a typical category?	A. The mean color is 2. B. The median is 2.5 colors. C. The standard deviation of color names is 1. D. Blue is the most common color. 10 pts
8. Two services have mean wait 10 minutes, but SDs 1 and 6 minutes. What does the larger SD suggest?	A. Waits vary more around the same mean. B. The median must be 6 minutes. C. Every wait is six minutes longer. D. The mean must actually be 16 minutes. 10 pts
9. A bell-shaped distribution of scores has mean 50 and standard deviation 5. Approximately what percentage of scores lies between 45 and 55?	A. 50%. B. 68%. C. 99.7%. D. 95%. 10 pts
10. A selected student's name is removed before the next random draw. What changes?	A. Selection has become a census. B. The same name must be selected next. C. That student cannot be selected again. D. Every remaining student becomes impossible to select. 10 pts

92081

TEST PAGE 1

1.

C.

10 pts

2.

D.

10 pts

3.

D.

10 pts

4.

D.

10 pts

5.

D.

10 pts

6.

C.

10 pts

TEST PAGE 2

7.

D.

10 pts

8. A.

10 pts

9.

B.

10 pts

10.

C.

10 pts

MSC 201 **50032**

Unit 1 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Ratings poor, fair, good, excellent are coded 1 through 4. What is safest to infer from the codes?	A. Excellent is exactly twice as favorable as fair. B. Excellent ranks above good, but equal numerical gaps are not established. C. Each adjacent rating differs by exactly one unit of satisfaction. D. The categories have no order. 10 pts
2. Notebook colors are blue, blue, black, red. Which statement meaningfully summarizes a typical category?	A. The mean color is 2. B. The standard deviation of color names is 1. C. Blue is the most common color. D. The median is 2.5 colors. 10 pts
3. Two datasets have identical five-number summaries. What follows?	A. Their means must be identical. B. Their sample sizes must be identical. C. They must contain identical observations. D. They can still have different internal distributions. 10 pts
4. A survey records whether each student uses a service. Which summary directly answers how common use is?	A. The mean student ID. B. The proportion of students reporting use. C. The largest student ID. D. The total of category labels chosen arbitrarily. 10 pts
5. In 2, 3, 3, 4, 18, the 18 is replaced by 180. What happens?	A. The mean increases; the median stays 3. B. The median increases more than the mean. C. Both stay unchanged. D. The mean decreases. 10 pts
6. An open online poll receives responses mainly from students strongly opposed to a fee. What is the concern?	A. All random samples favor strong opinions. B. The percentage becomes exact with 100 responses. C. Willingness to respond may depend on opinion. D. An online survey necessarily has no data. 10 pts

Test continues on the other side of the sheet.

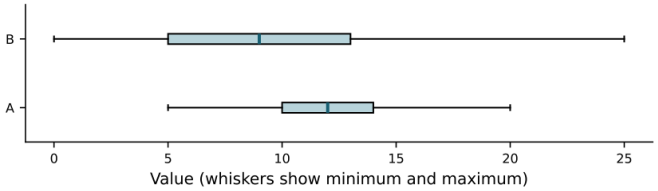
MSC 201 **50032**

Unit 1 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. Volunteers are randomly assigned to two practice formats. What does random assignment primarily help with?	A. Ensuring volunteers represent all students. B. Balancing other influences between treatment groups. C. Guaranteeing identical scores in both groups. D. Eliminating the need to measure outcomes. 10 pts
8. Box plot A has median 12, IQR 4; B has median 9, IQR 8. Which statement is supported? 	A. B has higher median and smaller spread. B. Every value in A exceeds every value in B. C. A has lower median and larger spread. D. A has higher median and smaller middle-half spread. 10 pts
9. Two stemplots use different keys: one leaf unit is 1 second, the other 10 seconds. What must happen before comparing spreads visually?	A. Assume equal stem lengths imply equal spread. B. Count only the number of stems. C. Delete repeated leaves. D. Account for the different numerical scales. 10 pts
10. A sample has 8 observations. Its sum of squared deviations is already supplied. Which divisor gives sample variance?	A. 64. B. The sample mean. C. 8. D. 7. 10 pts

50032

TEST PAGE 1

1.

B.

10 pts

2.

C.

10 pts

3.

D.

10 pts

4.

B.

10 pts

5. A.

10 pts

6.

C.

10 pts

TEST PAGE 2

7.

B.

10 pts

8.

D.

10 pts

9.

D.

10 pts

10.

D.

10 pts

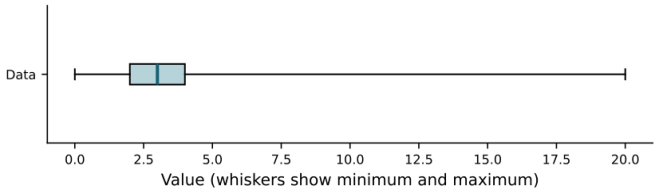
MSC 201 **72586**

Unit 1 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A recorded file size is far above the rest. What should happen before deleting it?	A. Delete it because all outliers are mistakes. B. Check whether it is an error or a genuine unusual file. C. Keep it only if the mean equals the median. D. Replace it with zero automatically. 10 pts
2. A dataset has mean 8. One observation below 8 is removed. What happens to the mean of the remaining values?	A. It always stays 8. B. It decreases. C. It becomes the removed value. D. It increases. 10 pts
3. A basic box plot reports minimum 0, Q1=2, median 3, Q3=4, maximum 20. Which comparison is supported? 	A. The segment from Q3 to the maximum spans a wider numerical range than the segment from the minimum to Q1. B. The upper segment must contain three times as many observations. C. The plot establishes the exact mean. D. The lower segment contains no observations. 10 pts
4. A grader does not know which practice format each paper's author used. What problem does this reduce?	A. Students being absent from the population. B. Expectations influencing grading. C. The possibility of random assignment. D. All variation in student ability. 10 pts
5. A library surveys only people entering the library to estimate how often all students visit. What is the main concern?	A. Randomly ordering the responses removes the concern. B. Students who never visit are excluded. C. The sample must be too small. D. Every recorded visit count is wrong. 10 pts
6. A frequency table lists 1 page for 8 jobs and 5 pages for 2 jobs. Why is (1+5)/2 inappropriate for mean pages per job?	A. It ignores how often each value occurs. B. It is always the median. C. It gives extra weight to 1-page jobs. D. It counts too many individual jobs. 10 pts

Test continues on the other side of the sheet.

MSC 201 **72586**

Unit 1 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. A cumulative table gives 30% below 5 seconds and 75% below 10 seconds. What percent is from 5 to less than 10?	A. 30%. B. 45%. C. 105%. D. 75%. 10 pts
8. A sample has 8 observations. Its sum of squared deviations is already supplied. Which divisor gives sample variance?	A. 64. B. The sample mean. C. 7. D. 8. 10 pts
9. A bar graph of counts 98 and 100 starts its vertical axis at 97. What impression might it exaggerate?	A. The fact that 98 is below 100. B. The number of categories. C. The size of the difference between the counts. D. The total number of observations. 10 pts
10. An open online poll receives responses mainly from students strongly opposed to a fee. What is the concern?	A. Willingness to respond may depend on opinion. B. The percentage becomes exact with 100 responses. C. An online survey necessarily has no data. D. All random samples favor strong opinions. 10 pts

72586

TEST PAGE 1

1.

B.

10 pts

2.

D.

10 pts

3. A.

10 pts

4.

B.

10 pts

5.

B.

10 pts

6. A.

10 pts

TEST PAGE 2

7.

B.

10 pts

8.

C.

10 pts

9.

C.

10 pts

10. A.

10 pts

MSC 201 **67706**

Unit 1 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A researcher deletes genuine results that disagree with the desired conclusion. What is the problem?	A. The report no longer fairly represents the collected evidence. B. Deleting inconvenient values always improves accuracy. C. Any unusual observation is automatically an error. D. A smaller dataset guarantees less bias. 10 pts
2. Plan A samples students from every building. Plan B randomly chooses two buildings and surveys everyone there. What distinguishes Plan B?	A. It studies all students in selected groups. B. It necessarily includes every building. C. It guarantees all possible student samples are equally likely. D. It selects some students from every group. 10 pts
3. Durations are in seconds. Which units belong to variance and standard deviation, respectively?	A. Percent; percent. B. Seconds squared; seconds. C. Seconds; seconds squared. D. Seconds; no units. 10 pts
4. Q1 is 8 and Q3 is 14. What does the interval 8 to 14 describe approximately?	A. The lowest quarter only. B. The mean's confidence interval. C. The middle half of observations. D. Every observation. 10 pts
5. A stemplot key says stem 2, leaf 5 means 25 KB. Stem 3 has leaves 0, 2, 2. Which values does it show?	A. 300, 320, 320 KB. B. 30, 32 KB only. C. 3, 0, 2, 2 KB. D. 30, 32, 32 KB. 10 pts
6. The numbers of pages in six jobs are 1, 2, 2, 4, 4, 5. Which statement about the mode is correct?	A. Only 2 is a mode. B. The mode is 3, halfway between 2 and 4. C. Only 4 is a mode. D. Both 2 and 4 are modes. 10 pts

Test continues on the other side of the sheet.

MSC 201 **67706**

Unit 1 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. A cumulative table gives 30% below 5 seconds and 75% below 10 seconds. What percent is from 5 to less than 10?	A. 30%. B. 75%. C. 45%. D. 105%. 10 pts
8. A recorded file size is far above the rest. What should happen before deleting it?	A. Keep it only if the mean equals the median. B. Delete it because all outliers are mistakes. C. Check whether it is an error or a genuine unusual file. D. Replace it with zero automatically. 10 pts
9. A sample variance is 9 minutes squared. What is its standard deviation?	A. 9 minutes. B. 3 minutes squared. C. 3 minutes. D. 81 minutes. 10 pts
10. A survey records whether each student uses a service. Which summary directly answers how common use is?	A. The mean student ID. B. The total of category labels chosen arbitrarily. C. The proportion of students reporting use. D. The largest student ID. 10 pts

67706

TEST PAGE 1

1. A.

10 pts

2. A.

10 pts

3. B.

10 pts

4. C.

10 pts

5. D.

10 pts

6. D.

10 pts

TEST PAGE 2

7. C.

10 pts

8. C.

10 pts

9. C.

10 pts

10. C.

10 pts

MSC 201 **76413**

Unit 1 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Histogram bins have unequal widths but equal counts. Why can equal raw-count bar heights mislead about concentration?	A. Equal counts require equal widths. B. The wider bin spreads its count over more numerical space. C. Every histogram must be a time series. D. Wider bins always contain smaller values. 10 pts
2. A well-run randomized experiment finds better results for an app among its volunteers. Which claim needs extra caution?	A. The observed volunteers had different average results. B. The study provides evidence about an app effect in this setting. C. Random assignment helped compare the formats. D. The same effect applies to all students everywhere. 10 pts
3. A room at 20 degrees Celsius is compared with one at 10 degrees Celsius. Which statement is justified?	A. The first is twice as comfortable for every person. B. The first is 10 degrees warmer on that scale. C. Celsius readings cannot be subtracted. D. The first has twice the absolute temperature. 10 pts
4. Compare 4, 5, 6 with 1, 5, 9. Which statement is correct?	A. They have equal means, but the second has more spread. B. The second has a larger mean. C. Equal means imply equal spread. D. The first has more spread. 10 pts
5. A database codes printer brands as 1, 2, and 3. Why would averaging these codes be misleading?	A. The largest code is always the most common brand. B. Brand codes cannot be counted. C. The numbers are labels; recoding them could change the mean. D. Every numerical entry measures an amount. 10 pts
6. Students are randomly invited, but only 15% respond. Which concern remains?	A. Low response proves every answer false. B. The invitations were necessarily nonrandom. C. Random invitation guarantees representative responses. D. Respondents may differ from nonrespondents. 10 pts

Test continues on the other side of the sheet.

MSC 201 **76413**

Unit 1 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. A dataset has mean 8. One observation below 8 is removed. What happens to the mean of the remaining values?	A. It increases. B. It becomes the removed value. C. It always stays 8. D. It decreases. 10 pts
8. Notebook colors are blue, blue, black, red. Which statement meaningfully summarizes a typical category?	A. The median is 2.5 colors. B. The mean color is 2. C. The standard deviation of color names is 1. D. Blue is the most common color. 10 pts
9. Using the reported quartiles, group A has Q1=3, Q3=9; group B has Q1=10, Q3=14. Which has greater IQR?	A. They are equal. B. Group A. C. Group B. D. The larger median must determine the IQR. 10 pts
10. In 2, 3, 3, 4, 18, the 18 is replaced by 180. What happens?	A. The median increases more than the mean. B. The mean increases; the median stays 3. C. The mean decreases. D. Both stay unchanged. 10 pts

76413

TEST PAGE 1

1.

B.

10 pts

2.

D.

10 pts

3.

B.

10 pts

4. A.

10 pts

5.

C.

10 pts

6.

D.

10 pts

TEST PAGE 2

7. A.

10 pts

8.

D.

10 pts

9.

B.

10 pts

10.

B.

10 pts

MSC 201 **23708**

Unit 1 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Service A has 10 complaints in 100 jobs; B has 20 in 400 jobs. Which has the lower complaint rate?	A. Service B. B. They have equal rates. C. Service A. D. The service with fewer complaints must have the lower rate. 10 pts
2. For 2, 3, 3, 4, 18, how does the mean compare with the median?	A. The mean is smaller. B. No comparison is possible without a calculator. C. The mean is larger. D. They are equal. 10 pts
3. A stemplot key says stem 2, leaf 5 means 25 KB. Stem 3 has leaves 0, 2, 2. Which values does it show?	A. 3, 0, 2, 2 KB. B. 30, 32 KB only. C. 300, 320, 320 KB. D. 30, 32, 32 KB. 10 pts
4. Campus A has 30 users among 60 students; campus B has 40 among 100. Which has the higher use rate?	A. Campus A. B. Campus B. C. Counts alone prove campus B has the higher rate. D. They have equal rates. 10 pts
5. A bell-shaped distribution of scores has mean 50 and standard deviation 5. Approximately what percentage of scores lies between 45 and 55?	A. 95%. B. 68%. C. 99.7%. D. 50%. 10 pts
6. A grader does not know which practice format each paper's author used. What problem does this reduce?	A. Expectations influencing grading. B. All variation in student ability. C. Students being absent from the population. D. The possibility of random assignment. 10 pts

Test continues on the other side of the sheet.

MSC 201 **23708**

Unit 1 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. Two reasonable quartile methods give upper outlier fences of 18 and 20 for the same dataset. One observation is 35. Which conclusion agrees with both methods?	A. The observation is not flagged under either method. B. The observation is flagged as unusually high under both methods. C. Only the method with fence 18 flags the observation. D. The observation must be a recording error. 10 pts
8. Students are randomly invited, but only 15% respond. Which concern remains?	A. Random invitation guarantees representative responses. B. Respondents may differ from nonrespondents. C. The invitations were necessarily nonrandom. D. Low response proves every answer false. 10 pts
9. Durations are in seconds. Which units belong to variance and standard deviation, respectively?	A. Seconds; no units. B. Seconds; seconds squared. C. Percent; percent. D. Seconds squared; seconds. 10 pts
10. Most scores cluster near the high end, with a few scores extending far toward smaller values. How is the distribution described?	A. Right-skewed. B. Symmetric. C. Uniform. D. Left-skewed. 10 pts

23708

TEST PAGE 1

1. A.

10 pts

2. C.

10 pts

3. D.

10 pts

4. A.

10 pts

5. B.

10 pts

6. A.

10 pts

TEST PAGE 2

7. B.

10 pts

8. B.

10 pts

9. D.

10 pts

10. D.

10 pts

MSC 201 **28976**

Unit 1 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Four mutually exclusive, exhaustive categories have reported proportions 0.20, 0.30, 0.40, and 0.25. What needs checking?	A. No category may exceed 0.20. B. Each category must have proportion 0.25. C. Their total exceeds 1. D. Proportions must be whole numbers. 10 pts
2. Four jobs have a mean of 5 pages. What is the total number of pages?	A. 9. B. 5. C. 1.25. D. 20. 10 pts
3. One teacher uses format A and another uses format B. Why is a score difference hard to attribute?	A. Teacher and format effects are mixed together. B. There are too many treatment groups to compare. C. Scores cannot be numerical. D. Different teachers guarantee format A is better. 10 pts
4. Datasets share min=0 and max=100, but have different IQRs. What follows?	A. Equal ranges do not imply equal middle-half spread. B. Their standard deviations must be identical. C. Their medians must be identical. D. Their observations must be identical. 10 pts
5. Remove 100 from 1, 2, 3, 4, 100. What happens to the mean and median?	A. The median increases and mean stays unchanged. B. Both stay unchanged. C. Both decrease, with a much larger decrease in the mean. D. The mean increases and median decreases. 10 pts
6. A random sample of 80 enrolled students reports 30% support for later hours. Which conclusion is justified?	A. 30% describes the sample and estimates population support. B. The population percentage must exceed 30%. C. Exactly 24 students in the entire college support later hours. D. Exactly 30% of all students support later hours. 10 pts

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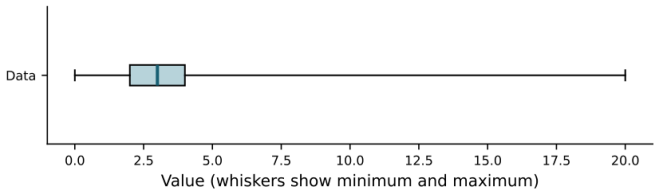
MSC 201 **28976**

Unit 1 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. Plan A samples students from every building. Plan B randomly chooses two buildings and surveys everyone there. What distinguishes Plan B?	A. It guarantees all possible student samples are equally likely. B. It studies all students in selected groups. C. It necessarily includes every building. D. It selects some students from every group. 10 pts
8. A basic box plot reports minimum 0, Q1=2, median 3, Q3=4, maximum 20. Which comparison is supported? 	A. The upper segment must contain three times as many observations. B. The segment from Q3 to the maximum spans a wider numerical range than the segment from the minimum to Q1. C. The plot establishes the exact mean. D. The lower segment contains no observations. 10 pts
9. Categories in a bar graph are reordered alphabetically. What changes?	A. The most frequent category. B. The number of observations in each category. C. Every relative frequency. D. The display order, but not category frequencies. 10 pts
10. Score A is 10 above its mean with SD 5; score B is 12 above its mean with SD 12. Which is relatively farther above its group mean?	A. B. B. They are equally far. C. The larger raw score difference always wins. D. A. 10 pts

28976

TEST PAGE 1

1.

C.

10 pts

2.

D.

10 pts

3. A.

10 pts

4. A.

10 pts

5.

C.

10 pts

6. A.

10 pts

TEST PAGE 2

7.

B.

10 pts

8.

B.

10 pts

9.

D.

10 pts

10.

D.

10 pts

MSC 201 **30280**

Unit 1 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A sample has 8 observations. Its sum of squared deviations is already supplied. Which divisor gives sample variance?	A. 64. B. The sample mean. C. 7. D. 8. 10 pts
2. A grouped table reports 12 durations from 5 to less than 10 seconds. Can it tell exactly how many equal 7 seconds?	A. Yes; none may equal 7. B. Yes; exactly half must equal 7. C. Yes; all 12 must equal 7. D. No; exact positions within the interval are hidden. 10 pts
3. A dataset has mean 10. One new observation equal to 10 is added. What happens to the mean?	A. It rises. B. It falls. C. It becomes zero. D. It stays 10. 10 pts
4. A printer study uses only one printer per paper type. Why is that weak evidence?	A. Replication would eliminate all possible errors. B. Two observations guarantee a representative result. C. Paper type cannot affect printer performance. D. Printer-specific differences may dominate the comparison. 10 pts
5. Two groups have the same median. Group A has Q1=6 and Q3=10; group B has Q1=3 and Q3=13. Which conclusion is supported?	A. Their equal medians require equal IQRs. B. Every value in group A is smaller than every value in group B. C. Group A has less spread in its middle half. D. Group B has less spread in its middle half. 10 pts
6. A systematic sample starts at ID 3 and takes every tenth ID. Which IDs appear first?	A. 3, 13, 23. B. 3, 10, 20. C. 10, 20, 30. D. 3, 6, 9. 10 pts

Test continues on the other side of the sheet.

MSC 201 **30280**

Unit 1 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. A random sample of 80 enrolled students reports 30% support for later hours. Which conclusion is justified?	A. Exactly 24 students in the entire college support later hours. B. The population percentage must exceed 30%. C. Exactly 30% of all students support later hours. D. 30% describes the sample and estimates population support. 10 pts
8. Data 4, 5, 6 become 4, 5, 60. What happens to spread?	A. Only the mean changes. B. Range and standard deviation both increase. C. Range decreases. D. Standard deviation becomes zero. 10 pts
9. Histogram bins are [0,5), [5,10), [10,15). Where does a duration of exactly 5 belong?	A. The second bin. B. None of the bins. C. Both first and second bins. D. The first bin. 10 pts
10. A score is reported at approximately the 90th percentile. What kind of information does that give?	A. Its position relative to other scores. B. The class mean being 90. C. That 90% of exam answers were correct. D. A score of exactly 90 points. 10 pts

30280

TEST PAGE 1

1.

C.

10 pts

2.

D.

10 pts

3.

D.

10 pts

4.

D.

10 pts

5.

C.

10 pts

6. A.

10 pts

TEST PAGE 2

7.

D.

10 pts

8.

B.

10 pts

9. A.

10 pts

10. A.

10 pts

MSC 201 **58614**

Unit 1 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Five print jobs contain 2, 3, 3, 5, and 7 pages. What is the mean number of pages per job?	A. 3. B. 4. C. 5. D. 20. 10 pts
2. Students rate a service while its manager watches. Compared with anonymous responses, what might happen?	A. Sample size automatically doubles. B. Students may avoid negative ratings. C. The manager's presence guarantees accuracy. D. Every rating must become negative. 10 pts
3. Two services have mean wait 10 minutes, but SDs 1 and 6 minutes. What does the larger SD suggest?	A. Waits vary more around the same mean. B. Every wait is six minutes longer. C. The median must be 6 minutes. D. The mean must actually be 16 minutes. 10 pts
4. A researcher deletes genuine results that disagree with the desired conclusion. What is the problem?	A. Deleting inconvenient values always improves accuracy. B. A smaller dataset guarantees less bias. C. The report no longer fairly represents the collected evidence. D. Any unusual observation is automatically an error. 10 pts
5. Four mutually exclusive, exhaustive categories have reported proportions 0.20, 0.30, 0.40, and 0.25. What needs checking?	A. Their total exceeds 1. B. No category may exceed 0.20. C. Each category must have proportion 0.25. D. Proportions must be whole numbers. 10 pts
6. Most scores cluster near the high end, with a few scores extending far toward smaller values. How is the distribution described?	A. Left-skewed. B. Right-skewed. C. Symmetric. D. Uniform. 10 pts

Test continues on the other side of the sheet.

MSC 201 **58614**

Unit 1 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. All ten observations equal 5, so their median is 5. Are exactly half strictly below the median?	A. No; a median cannot exist with ties. B. Yes; every median splits strict inequalities equally. C. Yes; exactly five are below 5. D. No; none are strictly below it. 10 pts
8. A box plot has median 6. Can it reveal exactly how many observations equal 5?	A. No; the summary does not retain those exact counts. B. Yes; none equal 5. C. Yes; half equal 5. D. Yes; exactly one quarter equal 5. 10 pts
9. Two stemplots use different keys: one leaf unit is 1 second, the other 10 seconds. What must happen before comparing spreads visually?	A. Account for the different numerical scales. B. Delete repeated leaves. C. Count only the number of stems. D. Assume equal stem lengths imply equal spread. 10 pts
10. Two properly drawn random samples give support rates of 42% and 47%. Which explanation is reasonable?	A. Different rates prove one survey was fabricated. B. Chance sampling variation can produce different rates. C. The larger percentage must be the true population rate. D. Random samples always have equal percentages. 10 pts

58614

TEST PAGE 1

1. B.

10 pts

2. B.

10 pts

3. A.

10 pts

4. C.

10 pts

5. A.

10 pts

6. A.

10 pts

TEST PAGE 2

7. D.

10 pts

8. A.

10 pts

9. A.

10 pts

10. B.

10 pts

MSC 201 **22639**

Unit 1 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A bar graph of counts 98 and 100 starts its vertical axis at 97. What impression might it exaggerate?	A. The fact that 98 is below 100. B. The total number of observations. C. The size of the difference between the counts. D. The number of categories. 10 pts
2. A sample variance is 9 minutes squared. What is its standard deviation?	A. 3 minutes squared. B. 9 minutes. C. 3 minutes. D. 81 minutes. 10 pts
3. A public dataset includes names beside individual survey answers. Which change best protects privacy while preserving group summaries?	A. Keep names and omit the study methods. B. Remove identifying details before sharing. C. Publish only the names. D. Replace every answer with the overall mean. 10 pts
4. A score is reported at approximately the 90th percentile. What kind of information does that give?	A. Its position relative to other scores. B. The class mean being 90. C. A score of exactly 90 points. D. That 90% of exam answers were correct. 10 pts
5. Which change best addresses a poll that includes only frequent library users?	A. Shuffle the existing responses. B. Broaden selection to include nonusers. C. Collect ten times as many frequent-user responses. D. Report more decimal places. 10 pts
6. A frequency table lists 1 page for 8 jobs and 5 pages for 2 jobs. Why is (1+5)/2 inappropriate for mean pages per job?	A. It counts too many individual jobs. B. It gives extra weight to 1-page jobs. C. It is always the median. D. It ignores how often each value occurs. 10 pts
7. Frequencies for 0, 1, 2, and 3 jobs are 2, 3, 4, and 1. How many sessions had at most 1 job?	A. 3. B. 9. C. 2. D. 5. 10 pts

Test continues on the other side of the sheet.

MSC 201 **22639**

Unit 1 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. Campus A has 30 users among 60 students; campus B has 40 among 100. Which has the higher use rate?	A. Campus B. B. Campus A. C. Counts alone prove campus B has the higher rate. D. They have equal rates. 10 pts
9. A recorded file size is far above the rest. What should happen before deleting it?	A. Replace it with zero automatically. B. Keep it only if the mean equals the median. C. Delete it because all outliers are mistakes. D. Check whether it is an error or a genuine unusual file. 10 pts
10. Why is adding signed deviations from the sample mean unsuitable as a spread measure?	A. The mean is always the largest value. B. It necessarily counts each value twice. C. Positive and negative deviations cancel. D. All deviations are positive. 10 pts

22639

TEST PAGE 1

1.

C.

10 pts

2.

C.

10 pts

3.

B.

10 pts

4. A.

10 pts

5.

B.

10 pts

6.

D.

10 pts

7.

D.

10 pts

TEST PAGE 2

8.

B.

10 pts

9.

D.

10 pts

10.

C.

10 pts

MSC 201 **22438**

Unit 1 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Data 4, 5, 6 become 4, 5, 60. What happens to spread?	A. Only the mean changes. B. Range decreases. C. Range and standard deviation both increase. D. Standard deviation becomes zero. 10 pts
2. Volunteers are randomly assigned to two practice formats. What does random assignment primarily help with?	A. Eliminating the need to measure outcomes. B. Guaranteeing identical scores in both groups. C. Balancing other influences between treatment groups. D. Ensuring volunteers represent all students. 10 pts
3. A modified box plot marks 30 as a separate high outlier. Its upper whisker ends at 18. What does 18 represent?	A. The largest observed value not flagged as an outlier. B. The maximum of all observations. C. The mean. D. The upper fence whether observed or not. 10 pts
4. Most files are small, with a few extremely large files. Which better describes a typical file's size?	A. The median. B. The maximum. C. The total size alone. D. The mean student ID. 10 pts
5. All ten observations equal 5, so their median is 5. Are exactly half strictly below the median?	A. No; none are strictly below it. B. Yes; every median splits strict inequalities equally. C. No; a median cannot exist with ties. D. Yes; exactly five are below 5. 10 pts
6. A college wants each academic program represented in a survey. Which plan best achieves that directly?	A. Select the first students entering one room. B. Survey only the largest program. C. Randomly sample students within every program. D. Invite only students who previously complained. 10 pts
7. Daily print totals are shown in a histogram instead of a time-series graph. What information is lost?	A. The approximate distribution of totals. B. The number of daily observations. C. The existence of high totals. D. Which total occurred on which date. 10 pts

Test continues on the other side of the sheet.

MSC 201 **22438**

Unit 1 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. A table has positive frequency at exactly 2 jobs. How do counts for fewer than 2 and at most 2 compare?	A. Fewer than 2 has the larger count. B. At most 2 has the larger count. C. Neither can be found from exact frequencies. D. They must be equal. 10 pts
9. A library surveys only people entering the library to estimate how often all students visit. What is the main concern?	A. Every recorded visit count is wrong. B. Students who never visit are excluded. C. Randomly ordering the responses removes the concern. D. The sample must be too small. 10 pts
10. Durations are in seconds. Which units belong to variance and standard deviation, respectively?	A. Percent; percent. B. Seconds; no units. C. Seconds squared; seconds. D. Seconds; seconds squared. 10 pts

22438

TEST PAGE 1

1.

C.

10 pts

2.

C.

10 pts

3. A.

10 pts

4. A.

10 pts

5. A.

10 pts

6.

C.

10 pts

7.

D.

10 pts

TEST PAGE 2

8.

B.

10 pts

9.

B.

10 pts

10.

C.

10 pts

MSC 201 **37240**

Unit 1 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Randomly assigned groups have slightly different initial mean scores. What follows?	A. The study must have used a census. B. The assignment was necessarily dishonest. C. Random assignment requires exact matching of every score. D. Chance imbalance can occur despite random assignment. 10 pts
2. A selected student's name is removed before the next random draw. What changes?	A. The same name must be selected next. B. Selection has become a census. C. Every remaining student becomes impossible to select. D. That student cannot be selected again. 10 pts
3. A population contains 4 observations with a sum of squared deviations from its mean of 36. What is the population variance?	A. 9. B. 3. C. 12. D. 36. 10 pts
4. Ten jobs average 2 pages; thirty jobs average 6 pages. Where is the combined mean?	A. Between 2 and 6, closer to 6. B. Below 2. C. Above 6. D. Exactly halfway at 4. 10 pts
5. Service A has 10 complaints in 100 jobs; B has 20 in 400 jobs. Which has the lower complaint rate?	A. Service A. B. The service with fewer complaints must have the lower rate. C. They have equal rates. D. Service B. 10 pts
6. A bell-shaped distribution has mean 20 and SD 2. Approximately which interval contains 95%?	A. 20 to 24. B. 16 to 24. C. 18 to 22. D. 0 to 20. 10 pts

Test continues on the other side of the sheet.

MSC 201 **37240**

Unit 1 Test

Name: _____ Date: _____

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Circle one response per question in the right column.

7. A sample's mean file size is 18 KB. The study aims to estimate the mean across all files on a server. What does 18 KB represent?	A. A guaranteed maximum file size. B. The size of every file on the server. C. An estimate based on the sampled files. D. The exact population mean regardless of selection. 10 pts
8. Using the reported quartiles, group A has Q1=3, Q3=9; group B has Q1=10, Q3=14. Which has greater IQR?	A. Group B. B. Group A. C. They are equal. D. The larger median must determine the IQR. 10 pts
9. A histogram is rebuilt using wider bins. What is likely lost?	A. The need for an axis scale. B. The total number of observations. C. Fine detail such as small gaps and local peaks. D. The numerical nature of the variable. 10 pts
10. Most scores cluster near the high end, with a few scores extending far toward smaller values. How is the distribution described?	A. Left-skewed. B. Uniform. C. Symmetric. D. Right-skewed. 10 pts

37240

TEST PAGE 1

1.

D.

10 pts

2.

D.

10 pts

3. A.

10 pts

4. A.

10 pts

5.

D.

10 pts

6.

B.

10 pts

TEST PAGE 2

7.

C.

10 pts

8.

B.

10 pts

9.

C.

10 pts

10. A.

10 pts

MSC 201 **38090**

Unit 1 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A timer reports durations to the nearest second. What follows about the underlying duration?	A. It may vary continuously despite rounded records. B. It cannot be less than one second. C. It must change only in whole seconds. D. It is a categorical label. 10 pts
2. Three reported percentages are 33.3%, 33.3%, and 33.3%. What most reasonably explains the 99.9% total?	A. The data must be fabricated. B. Overlapping categories are certain. C. Missing exactly one participant. D. Rounding. 10 pts
3. Digital purchases rise from 40 of 100 to 60 of 200. What happened?	A. Both count and share fell. B. The count fell, but the share rose. C. Both count and share rose. D. The count rose, but the share fell. 10 pts
4. Two stemplots use different keys: one leaf unit is 1 second, the other 10 seconds. What must happen before comparing spreads visually?	A. Delete repeated leaves. B. Count only the number of stems. C. Account for the different numerical scales. D. Assume equal stem lengths imply equal spread. 10 pts
5. Why is adding signed deviations from the sample mean unsuitable as a spread measure?	A. All deviations are positive. B. Positive and negative deviations cancel. C. It necessarily counts each value twice. D. The mean is always the largest value. 10 pts
6. A researcher deletes genuine results that disagree with the desired conclusion. What is the problem?	A. A smaller dataset guarantees less bias. B. Any unusual observation is automatically an error. C. Deleting inconvenient values always improves accuracy. D. The report no longer fairly represents the collected evidence. 10 pts

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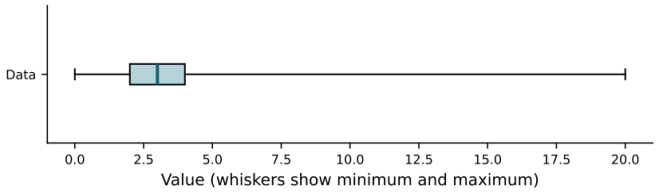
MSC 201 **38090**

Unit 1 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. A basic box plot reports minimum 0, Q1=2, median 3, Q3=4, maximum 20. Which comparison is supported? 	A. The segment from Q3 to the maximum spans a wider numerical range than the segment from the minimum to Q1. B. The plot establishes the exact mean. C. The lower segment contains no observations. D. The upper segment must contain three times as many observations. 10 pts
8. Five print jobs contain 2, 3, 3, 5, and 7 pages. What is the mean number of pages per job?	A. 5. B. 4. C. 3. D. 20. 10 pts
9. Most scores cluster near the high end, with a few scores extending far toward smaller values. How is the distribution described?	A. Left-skewed. B. Symmetric. C. Uniform. D. Right-skewed. 10 pts
10. Students are randomly invited, but only 15% respond. Which concern remains?	A. The invitations were necessarily nonrandom. B. Random invitation guarantees representative responses. C. Low response proves every answer false. D. Respondents may differ from nonrespondents. 10 pts

38090

TEST PAGE 1

1. A.

10 pts

2. D.

10 pts

3. D.

10 pts

4. C.

10 pts

5. B.

10 pts

6. D.

10 pts

TEST PAGE 2

7. A.

10 pts

8. B.

10 pts

9. A.

10 pts

10. D.

10 pts